Edgify Integration Guide - Linux & REST API

Overview

This document outlines the integration guide for integrating with Edgify via REST APIs on Linux machines. The Edgify Agent is installed using a container. The agent is launched on a linux machine using a docker-compose command and enables devices to execute real-time predictions. The Scale/POS/SCO UI application ("UI Application") interacts with the Edgify Agent ("Agent") via REST APIs.

High Level Overview of the Edgify API Calls:

Suggested UI Application Flow

Test Server with Edgify Agent

Additional Specifications

Detailed Guide for Edgify Rest API

- Prediction
- Set Ground Truth

Flow - Various Cases

- 1. Multiple Candidates
- 2. One Candidate
- 3. No Candidate ("Unknown" in Edgify Response)

High Level Overview of the Edgify API Calls:

- **Get Prediction** - For getting a list of probable identified items
- **Set Ground Truth** - For sending feedback to edgify with the item selected by the customer (for continuous learning)

Suggested UI Application Flow

1. **Scale Stable** - An Item is placed on the scale - Once Scale stabilizes, the UI application will call the Edgify Agent via REST API for **Get Prediction**
2. **Edgify Prediction** - The Edgify Agent will capture an image using the camera connected and respond with a prediction object (a variable-length array of suggested items, might be top 1 or top X).
3. **Candidates Selection -** The UI application should recommend to the customer the list provided by the Edgify Agent, if the prediction returns only 1 item, the UI application should skip the selection phase and continue directly to the barcode print phase.
4. **Barcode Label Print -** Once a customer confirms the prediction by printing the barcode label, the UI application will provide the Edgify Agent with the label and the last "get prediction" object using the **Set Ground Truth** API. The Edgify agent will redirect the photo to be saved on the Edgibox.

Additional Specifications

- Edgify agent to be installed in the scale or on an Edgibox (By Edgify)
- Edgify agent will take charge of the USB camera and recognition
- REST API to communicate with the Edgify Agent.
- The UI application should implement configurations for:
    - Agent URL
    - Agent port
    - Max timeout
    - Work Mode -
        - Edgify Off - No calls to Edgify
        - Learning Mode - Calling Edgify APIs but not showing candidate predictions.
        - Active Mode - Calling Edgify APIs and using Edgify prediction responses for showing predictions in the UI application

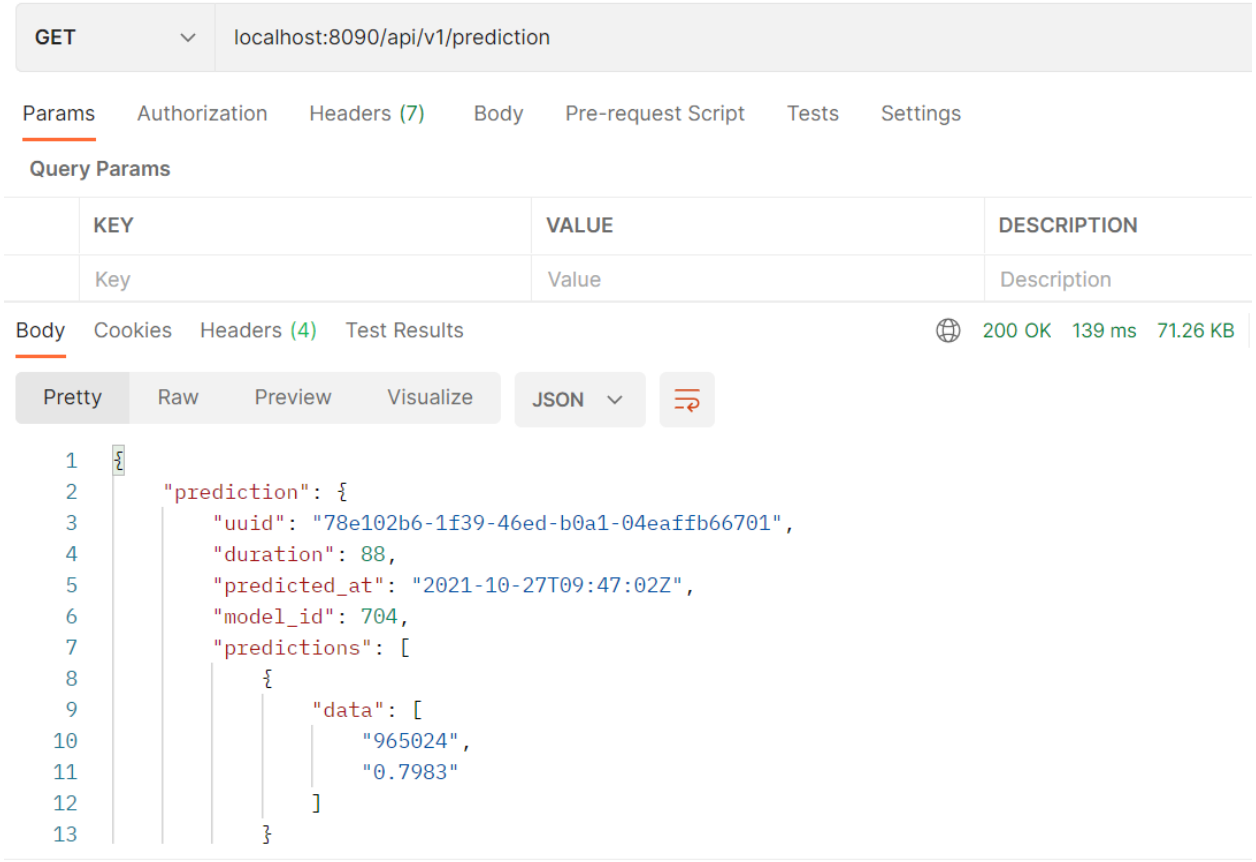
Detailed Guide for Edgify Rest API

Prediction

GET http://localhost:8090/api/v1/prediction

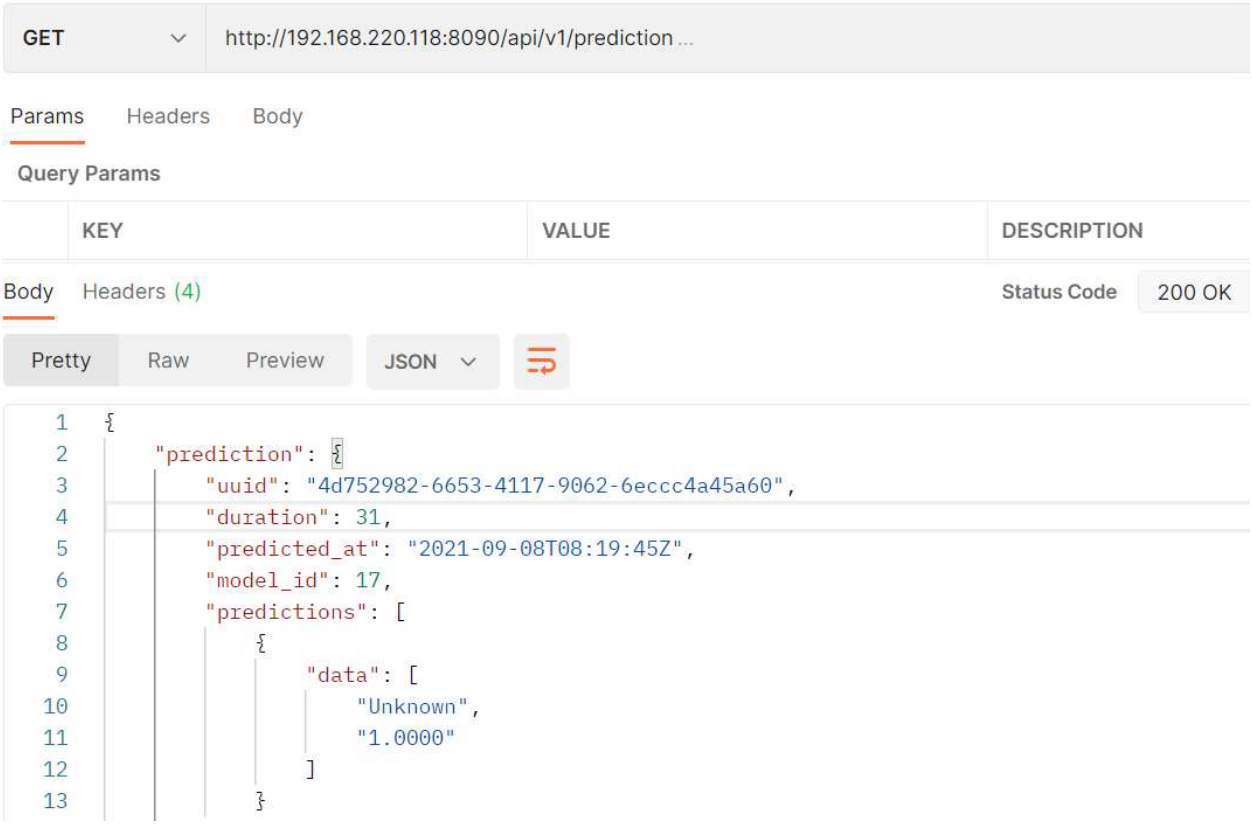
- Gets the predictions from the Agent, (UUID and the Image are in base64 text).

In the following example: The agent predicts the item is SKU # “965024” with Confidence of 0.7983:



- For unknown items, the agent will return a prediction object with “Unknown” in the prediction array.

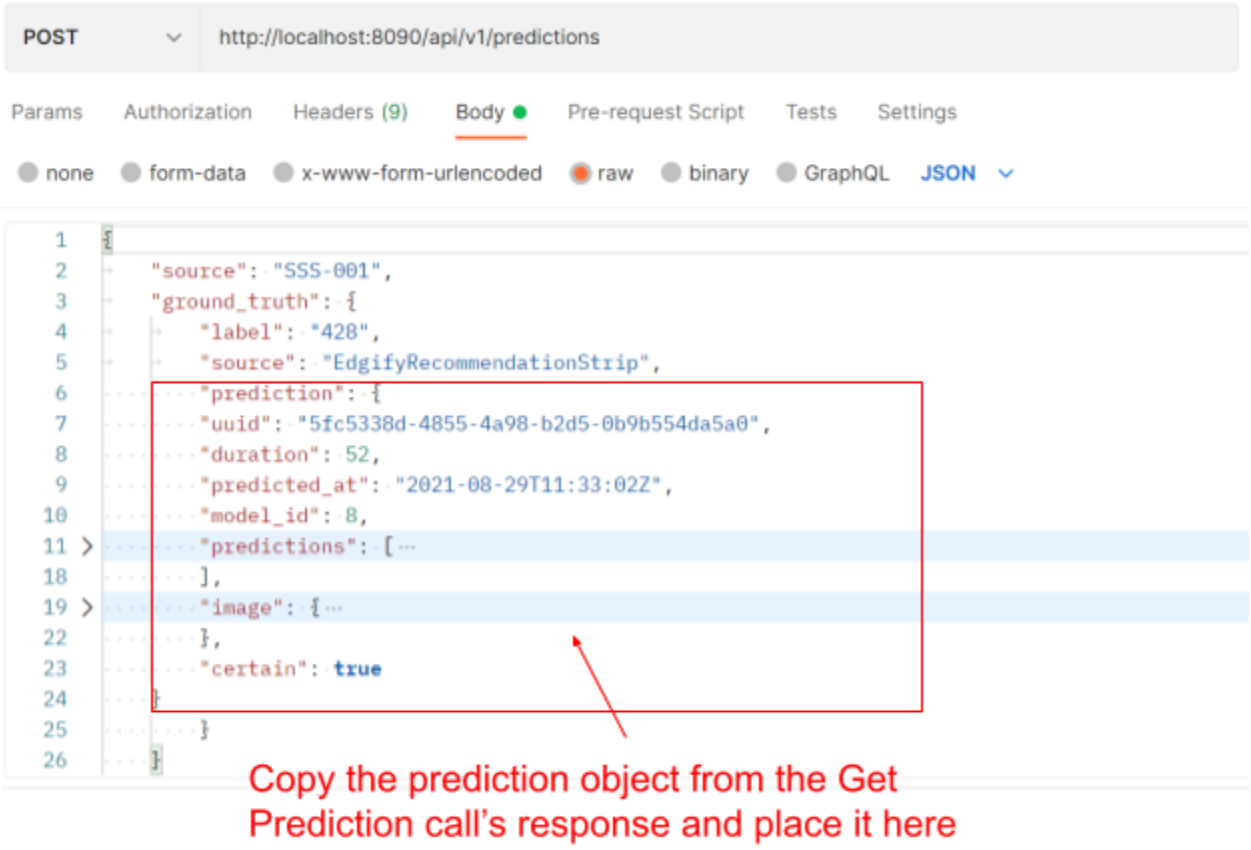
In the following example: The agent predicts the item is “Unknown”:



Set Ground Truth

POST http://localhost:8090/api/v1/predictions

- Saves the actual UUID and the Image with the chosen label, after the customer has chosen the correct product and printed the barcode label
- First “source”:”SSS-001” is the name of the device (scale). You can use the host name of the device running the UI application.
- “label”:”1005” is the product name /code
- Second “source”:”preset” can be any message, you should use this to identify where in the UX the customer chose to select the item to print.
  “detected item” or “by preset”
- “prediction: { uuid:……}”, is the prediction object from the last prediction. copy the response from GET Prediction



* Change “localhost” to Edgify Agent’s IP address of the test agent.

Flow - Various Cases

- 1. Multiple Candidates
  - 1. Place Item on platter
  - 2. Send GET Prediction once weight stable
  - 3. Scale displays all candidates
  - 4. Select PLU
  - 5. Print Label
  - 6. Send POST Set Ground Truth with selected PLU
- 2. One Candidate
  - 1. Place Item on platter
  - 2. Send GET Prediction once weight stable
  - 3. Scale auto call candidate
  - 4. Print Label
  - 5. Send POST Set Ground Truth with selected PLU
- 3. No Candidate (“Unknown” in Edgify Response)
  - 1. Place Item on platter
  - 2. Send GET Prediction once weight stable
  - 3. Show Preset Group
  - 4. Select PLU
  - 5. Print Label
  - 6. Send POST Set Ground Truth with selected PLU